GG-T-750

MARCH 4, 1958

SUPERSEDING
Int. Fed. Spec. GG-D-00796a (GSA-FSS)
March 1, 1956 and
Fed. Spec. GG-D-796
July 18, 1945

FEDERAL SPECIFICATION

TUBE, DUODENAL, SURGICAL, (REHFUSS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers duodenal outfits of Rehfuss design, used for draining the duodenum.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

GG-I-526 — Instruments, Dental and Surgical; General Specifications.

PPP-B-566 — Boxes, Folding, Paperboard.

PPP-B-585—Boxes; Wood, Wirebound.

PPP-B-601—Boxes; Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Test Method Std. No. 601—Rubber: Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-B-117—Bags, Interior, Packing.

MIL-B-10377 — Box; Wood, Cleated, Veneer, Paper Overlaid.

MIL-L-10547 — Liners, Case, Waterproof.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material.

3.1.1 *Rubber.*—The tube shall be of natural rubber, synthetic rubber or a mixture of the two. Color of the rubber shall be red or maroon. The rubber compound, or synthetic rubber compound, shall be free from any ingredient which may constitute a health hazard to personnel when used for the purpose intended.

3.1.2 *Metal.*—The metal shall comply with the applicable requirements for Class 4, Composition N of Federal Specification GG–I–526; and shall show no defects other than slight surface corrosion which will easily wash off, when tested as specified in section 4.

3.1.3 Other materials specified hereinafter shall be suitable for the intended purpose.

3.2 Description.—The complete duodenal outfit shall consist of a duodenal rubber tube and Rehfuss tip or bucket, and shall be equipped with an adapter (connector) and a rubber-syringe connecting tube unless the design of the outfit is such that these parts are unnecessary. A glass observation tube shall be included when specified in the invitation for bids.

3.3 Duodenal tube. — The duodenal tube shall be of one continuous piece of tubing 48 inches in length. It shall have an inside diameter of approximately ⅛ inch and a wall thickness of 1/32 inch (No. 14 French). The tube shall have one, two, or three encircling bands marked in permanent contrasting color, at points 16 inches, 27 inches, and 31 inches, respectively, when measured from the bucket end of the tube. Tubing shall have a smooth glass cured finish or other finish of equal smoothness.

3.4 Rehfuss tip or bucket.—The Rehfuss tip or bucket shall be made in one piece of metal as specified in 3.1.2. It shall be olive shape, measuring approximately ⅞ inch long over all, ⅜-inch-outside diameter, closed at one end and provided with a hollow tubular extension approximately ¼ inch long at the other end. The lumen of the hollow tubular extension shall be ⅛-inch diameter and communicate with the central opening of the bucket. The extension shall have a bulbous end, flared outward, or otherwise designed to offer maximum traction when inserted into the duodenal rubber tube. The bucket shall have four longitudinal slots communicating with a central opening, each approximately ½ inch long and 5/64 inch wide, located one in each quadrant.

3.5 Adapter (connector). — The adapter shall be of hard rubber or plastic. It shall be approximately 1 inch in length, with lumen equal to the lumen of the duodenal tube throughout its length. One end shall be of proper size to fit tightly into the duodenal rubber tube, the other end to fit tightly into the syringe connecting tube. The ends shall be bulbous, flare outward, corrugated or otherwise designed to offer maximum traction when inserted in the proper tube.

3.6 Syringe connecting tube.—The syringe connecting tube shall consist of one piece of rubber tubing 3 to 4½ inches in length with 1/16-inch wall thickness. One end of tube shall be of proper size to fit tightly over adapter, the other end to fit tightly over end of syringe.

3.7 Observation tube. — The observation tube shall be made of resistance glass, approximately 2½ inches in length and with a lumen equal to the lumen of the duodenal rubber tube. The ends shall be designed to provide a secure connection with the tubing. The observation tube shall be located between the syringe connecting tube and the duodenal tube.

3.8 Assembly.—Unless otherwise specified, the duodenal rubber tube shall be securely attached to the Rehfuss tip or bucket, by whipping with strong, undyed twisted silk thread. The thread shall be smoothly wound

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while wet and securely fastened by knotting the ends. The knot shall not extend beyond the widest portion of the bucket.

3.9 Identification marking. — Each tube shall be permanently and legibly marked with the name or registered trade-mark of the manufacturer, and date of manufacture (month and year).

3.10 Tolerances.—Unless otherwise noted, the following tolerances for dimensions shall be allowed:

Dimensions up to 1/16 inch	± 1/64 inch.
Dimensions over 1/16 inch up to 1/2 inch	± 1/32 inch.
Dimensions over 1/2 inch up to 1 inch	± 1/16 inch.
Dimensions over 1 inch up to 6 inches	± 1/8 inch.
All dimensions over 6 inches	± 1/2 inch.

3.11 Tensile strength. — The tensile strength shall be not less than 1,500 pounds per square inch.

3.12 Ultimate elongation.—The ultimate elongation shall be not less than 350 percent.

3.13 Resistance to accelerated aging.—The tensile strength and elongation of the tubing shall decrease not more than 25 percent when tested as described in 4.4.

3.14 Workmanship. — Workmanship shall be first class throughout. Duodenal outfits shall be free from defects which detract from their appearance or may impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for inspection. — Unless otherwise specified, inspection shall be in accordance with Military Standard MIL–STD–105. The acceptable quality level shall be 1.0 percent defective for major and 2.5 percent defective for minor.

4.2 Sampling for test.—Unless otherwise specified, the samples for test shall be taken as described in 4.2.3.2.1 of section 6 of Federal Test Method Standard No. 601. Sampling plan B of table II of section 6 of Federal Test Method Standard No. 601 shall be used.

4.3 Inspection.—Each unit in the sample for inspection, 4.1 shall be inspected for design, connections, marking, color, workmanship, and for dimensions.

4.4 Tests.

4.4.1 *Length.*—The length of the tube shall be determined as described in method 2411 of Federal Test Method Standard No. 601.

4.4.2 *Inside diameter.*—The average inside diameter shall be determined as described in method 2221 of Federal Test Method Standard No. 601.

4.4.3 *Wall thickness of tubing.*—The wall thickness of the tubing shall be determined by subtracting the inside diameter from the outside diameter at each place the inside and outside diameter is measured and dividing the result by 2. The smallest of all values obtained shall be the minimum wall thickness.

4.4.4. *Tensile strength.* — The tensile strength of the rubber compound shall be determined as described in method 4111 of Federal Test Method Standard No. 601 using a straight specimen.

4.4.5 *Ultimate elongation.*—The ultimate elongation shall be determined as described in method 4121 of Federal Test Method Standard No. 601 using a straight specimen.

4.4.6 *Resistance to accelerated aging.*—The resistance to accelerated aging shall be determined as described in method 7221 or 7231 of Federal Test Method Standard No. 601. The deterioration shall be measured by the change in tensile strength and elongation.

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4.4.7 *Salt-spray test.*—The salt-spray test shall be conducted as specified in Federal Specification GG–I–526.

4.4.8 *Boil test.*—The boil test shall be conducted as specified in Federal Specification GG–I–526.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Unit of issue.*—One tube, as specified, constitutes one unit.

5.1.2 *Quantities.*—The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

TABLE I.— *Packaging quantities*

Federal stock number	Unit package	Intermediate package	Shipping container
6515–386–0450	1 unit	12 units	360 units

5.1.2.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each unit shall be formed into a coil of such diameter that it will not kink or acquire a permanent set and shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Each carton shall be adequately secured to prevent accidental opening. Each carton shall be further packaged in a bag conforming to Military Specification MIL–B–117, type I, class d.

5.1.3.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level B (see section 6).*

5.1.4.1 *Unit package.*—Each unit shall be formed into a coil of such diameter that it will not kink or acquire a permanent set, and shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–566 or PPP–B–676. Each carton shall be adequately secured to prevent accidental opening.

5.1.4.2 *Intermediate package.*—Intermediate package shall be a carton of appropriate size and design, or a complete wrapping of kraft paper of suitable weight. Carton or wrapping shall be adequately secured to prevent accidental opening.

5.1.5 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and strapping shall conform to applicable paragraphs of appendix thereto.

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5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall conform to Federal Specification PPP–B–601, table I, using type III, class 1 plywood, or shall be designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 2 use; PPP–B–621, table I; or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Interior packages.*—Each unit carton and bag, and each intermediate package, shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2 *Shipping containers.*—Each shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein, and procurement documents should specify the following (see 3.8, 3.10, 4.1, 4.2 and 5.3.2):

(a) Title, number, and date of this specification.

(b) Observation tube, when required (see 3.2).

(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

(1) Level B packaging is intended to provide economical but limited protection, and should be specified only when it is determined that duodenal outfits will be held in covered storage approximately 12 months from date of initial packing.

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.5 Storage surveillance.—Items packaged in accordance with Level B requirements must be inspected to determine condition when not used within the time period indicated. Items not used within the time period specified must be repackaged in accordance with Level B requirements in this specification or with Level A requirements if storage beyond an additional year is anticipated.

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6.6 This specification covers the following item appearing in the Armed Services Medical Stock List:

6515–386–0450 — TUBE, DUODENAL, SURGICAL, Rehfuss, 14 Fr.

6.7 Transportation description.—Transportation description applicable to this item is:

Tubing, not otherwise indicated by name. Carload minimum weight 30,000 pounds. Truckload minimum weight 30,000 pounds.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for 5 cents each.

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